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Unveiling Customer Insights: Amazon Customers' Reviews Analysis with LDA and VADER

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Abstract

Online reviews show how customers see products and how satisfied they are. This paper combines topic modeling and sentiment analysis to find out what customers write about in Amazon reviews and how they feel about each theme. We use Latent Dirichlet Allocation (LDA) on a random sample of 10,000 reviews and add bigrams and trigrams, so that phrases of several words are not lost. We compare the coherence (C_v) of 24 models to choose the number of topics and then measure sentiment with the lexicon-based tool VADER. The model with the highest coherence has five topics (C_v = 0.539): books, movies, music, games and everyday use, and general products. All five topics have a positive average sentiment. Music is the most positive topic (mean compound score 0.56, 81% positive reviews) and movies are the least positive (0.29, 32% negative reviews). Coherence tends to fall when more topics are added, and several models are close to each other, so the number of topics should not be over-interpreted. VADER agrees with the star-based label of the dataset for about 70% of the reviews, so the sentiment per topic is only a rough guide. Even so, the approach shows how a company can find strengths and weaknesses in its customer feedback.

Keywords: Natural Language Processing, Topic Modeling, Sentiment Analysis, Amazon Reviews, LDA, VADER, Customer Feedback Analysis

Introduction

Customer reviews on e-commerce platforms like Amazon provide valuable insights into consumer experiences and opinions. However, the sheer volume of textual reviews makes it challenging for humans to manually digest and summarize the feedback. This has motivated the use of natural language processing techniques to automatically extract meaningful information from large collections of reviews. In particular, topic modeling and sentiment analysis are two complementary approaches that can help understand what customers are discussing and how they feel about those topics.

Topic modeling algorithms uncover latent thematic structure in text by grouping words into topics. A widely used method is Latent Dirichlet Allocation (LDA), a probabilistic model which assumes each document (here, a review) is a mixture of multiple topics and each topic is characterized by a distribution of words. Unlike simple clustering that assigns each document to one cluster, LDA allows documents to exhibit multiple themes, reflecting the reality that a single review might mention several aspects. Another important aspect of LDA is the usage of coherence score. It measures how interpretable the topics generated by the model are, based on the semantic similarity of the top words within each topic. LDA has been successfully applied in prior research to discover aspects of products that customers discuss in reviews. For example, Daravath et al. (2024) applied LDA to Amazon product reviews and demonstrated that it can uncover latent topics (themes) like product features or usage contexts, which can then be analyzed for their coherence and relevance.

Sentiment analysis, often called opinion mining, aims to determine the emotional tone or polarity (positive, negative, or neutral) of textual data. It has become an essential tool for businesses to gauge customer satisfaction and to extract opinions from online text. Approaches to sentiment analysis range from supervised machine learning classifiers to lexicon-based methods that use predefined sentiment dictionaries (e.g., lists of positive and negative words). In this study, we leverage VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon and rule-based sentiment analysis tool known for its effectiveness on social media and review text. VADER does not require training data and yields a compound score in the range of –1 (extremely negative) to +1 (extremely positive), with 0 indicating neutral sentiment. Its advantage is that it handles informal language, punctuation, and emoticons, making it well-suited for analyzing consumer-written reviews.

Objectives: This paper aims to combine LDA topic modeling with VADER sentiment analysis to analyze Amazon customer reviews. The primary goals are (1) to identify the main topics that occur in Amazon review texts and (2) to assess the sentiment associated with each discovered topic. By doing so, we seek to answer research questions such as “What are the prevalent themes in Amazon customer reviews?” and “Which topics are associated with highly positive or negative customer sentiment?” Understanding these can help businesses recognize what aspects customers praise or complain about most, thereby informing product improvements and customer service strategies.

Literature Review & Theoretical Background

Topic modeling has become a popular technique for extracting structured information from unstructured text like product reviews. The foundational LDA model by Blei, Ng, and Jordan (2003)

introduced the idea of uncovering latent topics in document collections through a generative probabilistic framework. Since then, numerous studies have applied LDA and its variants to consumer reviews to identify commonly discussed aspects of products. For example, Hu and Liu (2004) performed one of the early works on extracting product features (aspects) from customer reviews, while more recent approaches use LDA to automatically learn such aspects. Titov and McDonald (2008) showed that topic models can be tailored to focus on aspects and sentiments in reviews of hotels and restaurants. Ye and Srivastava (2019) specifically used LDA on ~400,000 Amazon mobile phone reviews to cluster words into aspect topics and then analyzed sentiment for each aspect, illustrating the value of combining topic modeling with aspect-specific sentiment analysis. These applications demonstrate that topic modeling helps summarize the main themes in thousands of reviews, enabling downstream analysis of each theme.

A key challenge in topic modeling is determining the appropriate number of topics and ensuring the topics are meaningful. Researchers often use topic coherence measures to quantitatively evaluate the quality of topics. Topic coherence measures how semantically related the top words of a topic are. A higher coherence score generally indicates a more interpretable topic where the top terms make sense together. Various coherence metrics have been proposed (e.g., UMass, UCI, and CV coherence), and studies have shown they correlate with human judgments of topic quality. Röder, Both, and Hinneburg (2015) introduced a framework for exploring different coherence measures and found that combining pointwise mutual information with context vectors (the CV coherence measure) yields robust evaluations of topic quality. In practice, coherence scores allow modelers to compare LDA models with different numbers of topics or preprocessing strategies to select the model that produces the most coherent topics.

Another consideration in applying LDA to reviews is incorporating n-grams (multi-word phrases). Standard LDA treats the text as a “bag of words,” which can break apart meaningful phrases (e.g., “customer service”) into separate words. Including bigrams or trigrams as tokens can help capture multi-word concepts that represent specific aspects more clearly. For instance, the phrase “highly recommended” or “shipping cost” might be an important compound term in reviews. Prior work has found that topic models incorporating such n-grams can yield more interpretable topics when analyzing domain-specific corpora like product reviews.

Sentiment analysis has been extensively studied in natural language processing, with surveys by Pang and Lee (2008) and Mukherjee and Liu (2012) outlining the evolution of the field. Early methods relied on supervised machine learning applied to labeled corpora of reviews (e.g., classifying movie reviews as positive or negative). As an alternative, lexicon-based methods use curated dictionaries of sentiment-laden words to score text. One popular lexicon method, VADER, was introduced by Hutto and Gilbert (2014) specifically for social media contexts. VADER provides not only a compound sentiment score but also classifies text as positive, neutral, or negative based on thresholding the compound score. It is effective for short, informal texts and accounts for things like capitalization (for emphasis), punctuation (e.g., “!!!” increases intensity), and emoticons.

Another example is the Joint Sentiment-Topic (JST) model by Lin and He (2009), which extends LDA by associating latent sentiment labels with topics within the generative process. JST and related models (e.g., aspect-based sentiment models) enable discovering topics and sentiment simultaneously. However, these models can be complex and computationally intensive. In practical analysis, a simpler pipeline combining separate topic modeling and sentiment analysis steps can be

effective. Our approach follows this simpler pipeline: first use LDA to identify topics, then apply sentiment analysis (VADER) to gauge the sentiment of each review (and by aggregation, each topic). This two-step approach has been used in several studies on reviews, for example, Daravath et al. (2024), yielding interpretable results that tie customer sentiment to specific themes.

Research Gaps and Contributions

Despite extensive research in topic modeling and sentiment analysis, several gaps remain in the literature. First, limited studies have systematically explored the impact of preprocessing techniques, particularly n-gram integration, on topic model performance in customer review analysis. Second, few studies have conducted comprehensive hyperparameter optimization for LDA models applied to large-scale review datasets. Third, the evaluation of different coherence measures for topic quality assessment in e-commerce contexts remains underexplored.

This research addresses these gaps by providing a systematic evaluation of LDA model configurations, comprehensive preprocessing pipeline optimization, and detailed analysis of topic-sentiment relationships in Amazon reviews. Our contribution extends beyond previous work by demonstrating the practical application of optimized topic modeling approaches to generate actionable business insights from customer feedback.

Methodology & Data

Dataset

We use the Amazon Review Polarity Dataset (Zhang, Zhao & LeCun, 2015), which was built from the SNAP Amazon reviews collection (McAuley & Leskovec, 2013). The underlying collection has about 35 million reviews from 18 years, up to March 2013. The polarity dataset keeps only reviews with 1–2 stars (negative) and 4–5 stars (positive). Its training file has 3.6 million reviews, 1.8 million per class. It is hard to model this much text, so we took a random sample of 10,000 reviews from the training file (random seed 42). One review was empty after the vocabulary filtering described below, so 9,999 reviews remain. The sample is balanced (50.3% negative, 49.7% positive) and covers different product types, for example books, music, movies, electronics and games.

The median review has 66 words. The language is informal, with slang, abbreviations and many exclamation marks, which is typical for text written by customers.

Text Preprocessing

We cleaned every review in the following steps:

- **Lowercasing:** all text is converted to lowercase, so “Great” and “great” count as the same word. The line breaks in the data (the characters `\n`) are replaced by spaces.
- **Tokenization:** the reviews are split into words with the NLTK word tokenizer.
- **Punctuation and numbers:** we keep only alphabetic tokens.
- **Stop words:** we remove the English stop words from NLTK and also a short list of 19 generic words (*one, like, would, could, get, got, also, even, much, really, good, great, well, thing, make, ever, still, back, way*). These words appear in almost every topic and make the topics look alike. Words like “good” and “great” are removed only for the topic model. VADER uses the original text.
- **Lemmatization:** the WordNet lemmatizer reduces words to their base form, for example “guitars” to “guitar”.
- **Bigrams and trigrams:** Gensim’s *Phrases* (minimum count 5, threshold 100) joins words that often appear together into one token. In our sample it found 206 different phrases, for example *waste_money* (176 times), *highly_recommend* (150), *main_character* (93), *customer_service* (72), *stopped_working* (47) and *poorly_written* (37).
- **Dictionary and corpus:** a Gensim dictionary gives every token an ID. Tokens that appear in fewer than 5 reviews or in more than 50% of the reviews are removed. This leaves 6,410 tokens (6,554 with n-grams). Each review is then stored as a bag of words with the token counts (*doc2bow*).

LDA works with word counts, so we use the raw counts for all models. The same corpus is used for the training, the coherence score, the visualization and the assignment of reviews to topics.

LDA Topic Modeling

We use LDA to find hidden topics in the reviews. LDA is a probabilistic model. It assumes that every document (here a review) is a mixture of topics and that every topic is a distribution over words. Words that often appear together in reviews end up in the same topic. We use the Gensim library (Řehůřek & Sojka, 2010), version 4.3.3. Gensim estimates LDA with online variational Bayes (Hoffman, Bach & Blei, 2010) and not with Gibbs sampling. All models are trained with the single-process *LdaModel*, 10 passes over the corpus and a fixed random seed (42), so the results can be repeated with the same software versions.

Model variants. We change three settings:

1. The number of topics K: 5, 8, 10, 15, 20 and 30.
2. α: the Dirichlet prior for the topics of a document.
3. η (also called β): the Dirichlet prior for the words of a topic.

For every K we train one model on the unigram corpus and one on the corpus with bigrams and trigrams, both with symmetric priors (12 models). Then we change the priors on the n-gram corpus for K = 5, 10 and 15 (12 more models): asymmetric α, learned α ("auto"), and α = η = 0.01 and α = η = 10. We do not learn η automatically, because in Gensim's *LdaModel* this option does not change the topics, so it would only repeat runs. In total we compare 24 models.

Model selection. We evaluate every model with the C_v coherence score. It measures how similar in meaning the top words of a topic are, and the score of a model is the average over its topics. A higher score means that the top words of a topic often appear together in the same reviews, which makes the topic easier to understand.

Table A1 in the appendix shows the coherence of all models. With symmetric priors, the five-topic model with n-grams has the highest coherence (0.539). The eight-topic unigram model (0.526) and the eight-topic n-gram model (0.520) follow. After that, coherence mostly falls when we add topics. With 15 topics it is 0.467 (unigrams) and 0.462 (n-grams), and with 30 topics 0.426 and 0.415. N-grams therefore do not always help: coherence is higher with n-grams for K = 5 and K = 20 but lower for K = 8, 10, 15 and 30. The other prior settings do not beat the symmetric model for K = 5 (learned α: 0.536), and very large priors (α = η = 10) give clearly worse topics (about 0.27 to 0.30).

We choose the five-topic model with n-grams because it has the highest coherence. Two points should be kept in mind. First, differences of a few hundredths are small compared with the change another random seed could cause, so the order of the best models (K = 5 and 8) should not be over-interpreted. Second, five topics is a coarse solution. It separates the big groups (books, movies, music) well, but it puts smaller themes together in wider clusters. Readers who want more detail can reproduce models with more topics with the notebook.

After choosing the model, we look at the top words of each topic and at the three reviews with the highest probability for that topic. Based on these, we give each topic a short label. For example, the topic with the words "album, cd, song, music, band" is labeled "Music", and the topic with "book, read, story, author, character" is labeled "Books & Reading". This step is subjective.

For every review, LDA gives a probability for each topic. The dominant topic of a review is the topic with the highest probability. We use it to assign each review to one topic for the sentiment analysis.

Sentiment Analysis with VADER

For the sentiment analysis we use VADER (Hutto & Gilbert, 2014) to compute a score for every review. VADER was made for social media and consumer text, which is similar to Amazon reviews because it is informal and often uses exclamation marks. We chose a lexicon-based method instead of a machine learning classifier, because it needs no labeled training data and its scores are easy to interpret. We proceed as follows:

1. **Text:** VADER is applied to the original review text and not to the lemmatized text from the LDA step, because it uses punctuation and capitalization to judge how strong a sentiment is.
2. **Scores:** VADER gives a positive, a neutral and a negative share and a compound score between −1 and +1. We use the compound score.
3. **Categories:** a review is positive if the compound score is at least 0.05, negative if it is at most −0.05 and neutral in between. These are the thresholds recommended by the authors of VADER.
4. **Topics:** for each dominant topic we compute the number of reviews, the mean compound score and the share of positive, neutral and negative reviews.
5. **Check:** the dataset labels each review as positive or negative based on its star rating. We compare VADER with this label.

Across the 9,999 reviews, VADER classifies 70.8% as positive, 25.8% as negative and 3.5% as neutral. A positive majority is typical for Amazon reviews. Note that the sample itself is balanced (50.3% / 49.7%) according to the star-based labels, so VADER rates more reviews as positive than the star ratings do.

Results

Discovered Topics from Amazon Reviews

The selected five-topic model with n-grams gives five topics. Table 1 shows their size, sentiment and top keywords.

Topic	Label	Reviews	Mean compound	Pos / Neu / Neg	Top keywords	One-line takeaway
0	Books & Reading	3,017	0.412	73% / 3% / 24%	book, read, story, author, reading, time, life, many, character, people	Readers discuss plots, authors and characters; mostly positive, but critical non-fiction reviews appear too.
1	Movies & Film	1,682	0.286	66% / 3% / 32%	movie, film, time, see, watch, love, dvd, show, bad, first	Viewers rate films and DVDs; praise (“love”) and disappointment (“bad”) sit side by side.
2	Games, Software & Everyday Use	968	0.359	70% / 5% / 25%	game, use, play, work, better, buy, need, time, problem, computer	Mixed cluster: computer and video games plus general “use / work / problem” reviews (examples include game and coffee reviews).
3	Music	1,523	0.558	81% / 2% / 16%	album, cd, song, music, sound, love, best, fan, dvd, band	Albums, songs and bands; the most positive topic. A few reviews are in Spanish.
4	General Products & Purchases	2,809	0.298	66% / 5% / 30%	product, work, bought, use, time, used, amazon, buy, price, quality	Product use, price and quality; 30% negative reviews (e.g. parts that broke).

Table 1: The five topics of the selected model (highest coherence, C_v = 0.539). “Pos / Neu / Neg” is the share of reviews that VADER classifies as positive, neutral and negative. The labels and takeaways are our interpretation of the keywords and example reviews.

Three topics have a clear focus (books, movies, music) and two are broader (games and everyday use; general products). Topic 2 is the least clear. Its three most typical reviews are about two video games and a coffee product, so it seems to collect reviews about use and function from several areas. One of the three most typical music reviews is written in Spanish, so a few non-English reviews end up in the music topic.

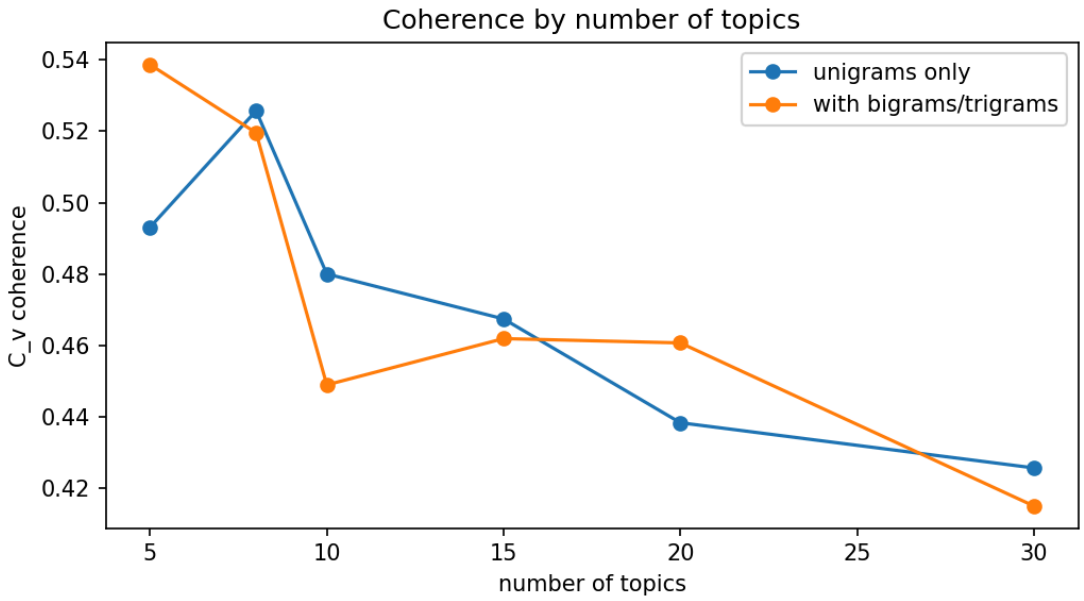


Illustration 1: Topic coherence (C_v) for different numbers of topics, with and without n-grams (symmetric priors, seed 42).

Sentiment Analysis Results

All five topics have a positive mean compound score. Music (topic 3) is the most positive (mean 0.558, 81% positive and 16% negative reviews). Movies (topic 1) have the lowest mean (0.286, 66% positive and 32% negative reviews), closely followed by general products (topic 4, 0.298, 30% negative). Books (0.412) and games and everyday use (0.359) are in between. Music clearly stands out from the other topics, while the three lowest topics (0.286 to 0.359) are close to each other.

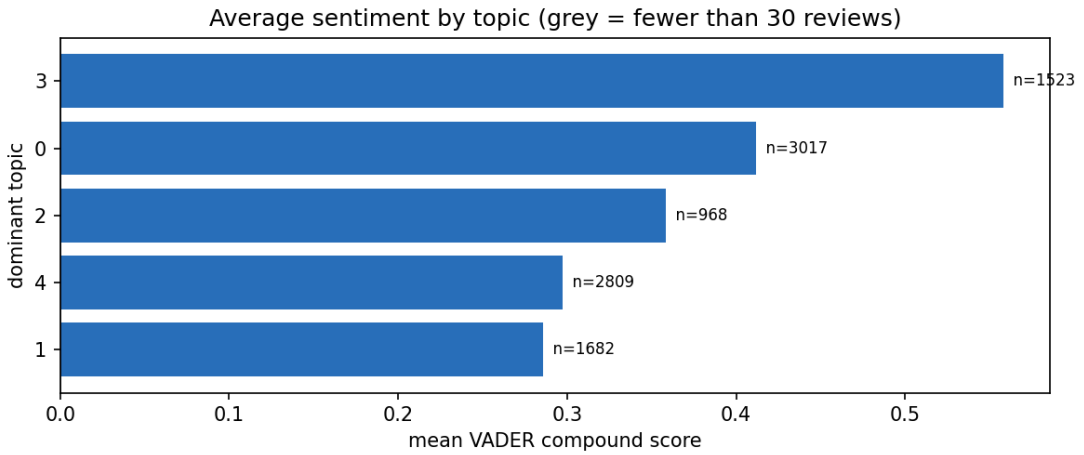


Illustration 2: Average VADER compound score by dominant topic (n = number of reviews).

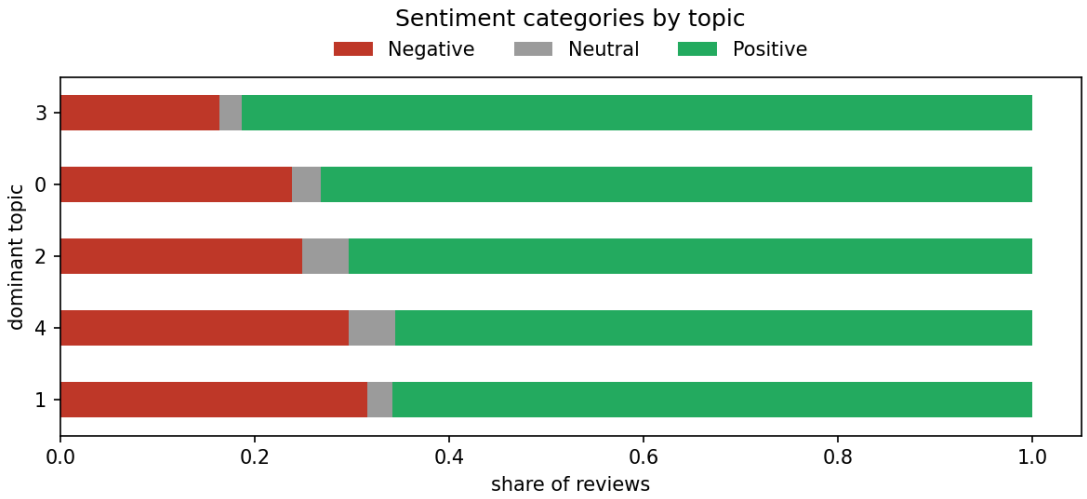


Illustration 3: Share of negative, neutral and positive reviews by dominant topic.

How reliable is the sentiment score? We compare VADER with the star-based label of the dataset. VADER is neutral for 3.5% of the reviews, and for the rest it agrees with the label in 70.4% of the cases. Most errors are negative reviews that VADER calls positive: 2,531 of 5,027 negative reviews are classified as positive, but only 326 of 4,972 positive reviews are classified as negative. The averages per topic should therefore be used as a rough comparison between topics and not as exact measurements. A topic can look too positive when reviewers write polite or sarcastic complaints with positive words.

Key Business Implications

For managers, the results can help in the following ways.

Marketing and communication: positive topics show selling points that can be emphasized. In our data the music reviews are especially positive, so product descriptions and advertisements for music can build on this enthusiasm. In the same way, the common complaints in a topic show which concerns a company should address.

Customer segmentation: the topics correspond to different product lines and customer groups. The share of negative reviews is clearly higher for movies (32%) and general products (30%) than for music (16%), so improvement work could start with these categories.

Monitoring: the method can be repeated regularly on new reviews. If the negative sentiment in a topic suddenly rises, this can be an early sign of a problem, for example a defective batch of products, and the company can react quickly.

Context: the method gives more information than star ratings or an overall sentiment score alone, because it shows in which areas customers are happy or unhappy. For decisions about single products, a finer topic resolution than five topics would be needed.

Limitations

Sample and data.

- 10,000 reviews are only a small part of the 3.6 million reviews in the polarity dataset.
- The reviews come from mixed product categories and were written up to 2013, so changes over time and newer products are not covered.
- The results depend on the random sample and on the random seed of the LDA training. All numbers are for seed 42, and other seeds can change topics and coherence scores a little.

LDA model.

- LDA assumes clear topic boundaries, but customer concerns overlap. The five-topic solution is coarse, and one topic (games and everyday use) is mixed.
- Coherence is only one criterion. Here it favored few topics, and the best models are close to each other.
- The dominant topic ignores that a review can talk about several topics.

VADER.

- VADER has problems with sarcasm and with special words that are not in its lexicon.
- It agrees with the star-based label for only about 70% of the non-neutral reviews, and it more often calls a negative review positive than the other way round.
- One compound score can hide small differences in feelings.

Other points.

- The phrase detection finds useful phrases (for example *customer_service* and *waste_money*), but with our thresholds it also joins some pairs that say little (for example *year_ago* and *ca_wait*).
- One person labeled all topics, so personal bias is possible.
- Reviews in other languages (mostly Spanish) were not removed.

Future Work & Conclusion

Future work: there are several ways to continue this study:

- **Other topic models:** try methods like Non-Negative Matrix Factorization (NMF) or neural topic models.
- **More data:** use a larger sample or look at single product categories. This would allow more topics.
- **Stability:** train the models with several random seeds and report how much the coherence scores change.
- **Ratings and metadata:** compare the sentiment per topic with star ratings or sales data, and compare VADER with a supervised classifier.

- **Manual check:** read reviews from the most and the least positive topics to see whether the interpretation fits (possibly with several raters).

Conclusion: In this paper we combined LDA topic modeling and VADER sentiment analysis to study Amazon reviews. On a sample of 10,000 reviews, the model with the highest coherence has five topics: books, movies, music, games and everyday use, and general products. The sentiment is positive on average in all topics, highest for music and lowest for movies, and the share of negative reviews goes from 16% to 32%. Two findings limit these results. The best models are close in coherence and n-grams did not improve it everywhere, and VADER agrees with the star-based label for only about 70% of the reviews. Still, the two-step approach shows more than either method alone, because it tells us what customers write about and how the sentiment differs between these themes. With the fixed random seed and the notebook, the analysis can be repeated and extended to larger samples and more topics.

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Appendix

Appendix A: Coherence of all models

Table A1: Models with symmetric priors (α = η = symmetric), with and without n-grams.

Corpus	# Topics	N-grams?	α	η	Coherence (C_v)
Unigram tokens	5	No	symmetric	symmetric	0.493
Unigram tokens	8	No	symmetric	symmetric	0.526
Unigram tokens	10	No	symmetric	symmetric	0.480
Unigram tokens	15	No	symmetric	symmetric	0.467
Unigram tokens	20	No	symmetric	symmetric	0.438
Unigram tokens	30	No	symmetric	symmetric	0.426
Bigrams/trigrams	5	Yes	symmetric	symmetric	0.539
Bigrams/trigrams	8	Yes	symmetric	symmetric	0.520
Bigrams/trigrams	10	Yes	symmetric	symmetric	0.449
Bigrams/trigrams	15	Yes	symmetric	symmetric	0.462
Bigrams/trigrams	20	Yes	symmetric	symmetric	0.461
Bigrams/trigrams	30	Yes	symmetric	symmetric	0.415

Table A2: Prior search on the n-gram corpus for K = 5, 10 and 15 (other settings than symmetric/symmetric).

Corpus	# Topics	N-grams?	α	η	Coherence (C_v)
Bigrams/trigrams	5	Yes	auto	symmetric	0.536
Bigrams/trigrams	5	Yes	0.01	0.01	0.514
Bigrams/trigrams	5	Yes	asymmetric	symmetric	0.502
Bigrams/trigrams	5	Yes	10	10	0.298
Bigrams/trigrams	10	Yes	asymmetric	symmetric	0.456
Bigrams/trigrams	10	Yes	0.01	0.01	0.438
Bigrams/trigrams	10	Yes	auto	symmetric	0.429
Bigrams/trigrams	10	Yes	10	10	0.274
Bigrams/trigrams	15	Yes	asymmetric	symmetric	0.471
Bigrams/trigrams	15	Yes	auto	symmetric	0.458
Bigrams/trigrams	15	Yes	0.01	0.01	0.456
Bigrams/trigrams	15	Yes	10	10	0.273